



Navigation

- Select View:
- 1. Executive Overview & Risk Tiers
 - 2. Model Evaluation & Dual PR-AUC
 - 3. Feature Drivers & Interpretability
 - 4. Customer Explorer & Action Center

Methodology Specifications

- Snapshot Date: 2018-02-28
- Target Horizon: 180 days
- Window: 2018-03-01 to 2018-08-27
- Customer Key: customer_unique_id
- High-Risk Threshold: > 70%
- Medium-Risk: 30% - 70%
- Low-Risk: < 30%

Methodological Architecture

Deploy

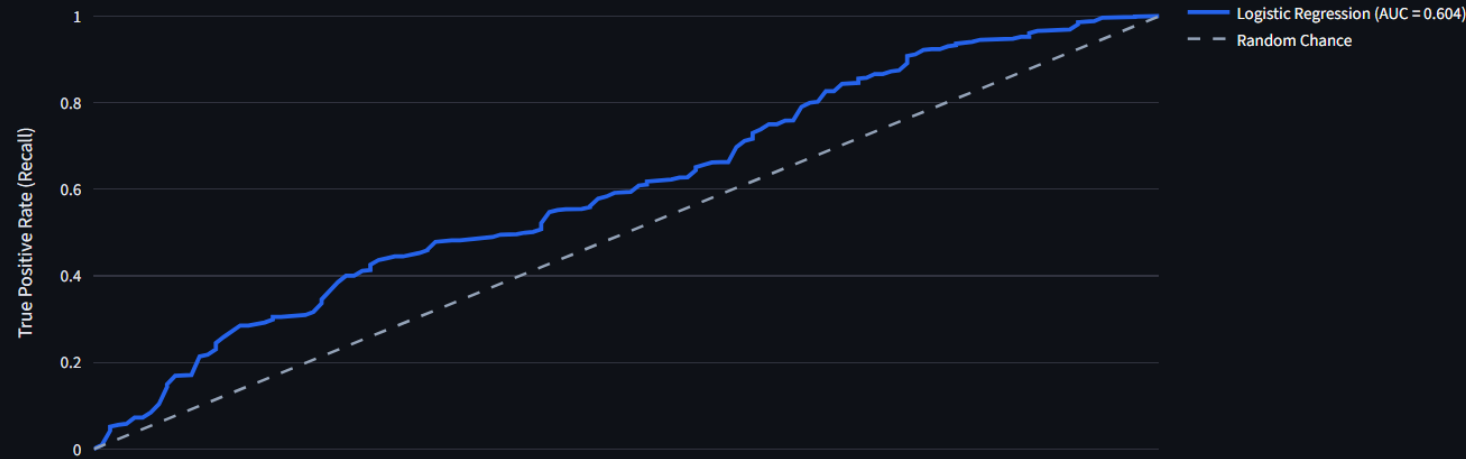
Model Performance & Dual PR-AUC Laboratory

Rigorous statistical evaluation under extreme class imbalance (98.8% Churn vs. 1.18% Active).

Test ROC-AUC	PR-AUC (Churn=1)	PR-AUC (Active=0)	Brier Score
0.6035	0.9918	0.0312	0.011557
	↑ Baseline: 0.9882	↑ Baseline: 0.0118	↑ Calibrated (Lower=Better)

ROC Curve Dual PR-AUC Curves Confusion Matrix Decile Lift & Cumulative Gains

Receiver Operating Characteristic (ROC) Curve



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Feature Drivers & Behavioral Coefficients

Transparent Logistic Regression Odds Ratios and churn risk sensitivities.

How to Interpret Odds Ratios:

- **Odds Ratio > 1.0 (Red):** Increasing this feature multiplies the odds of customer churn.
- **Odds Ratio < 1.0 (Green):** Increasing this feature reduces churn odds (promotes customer retention).
- **Baseline Value (1.0):** Neutral effect on churn probability.

Top Insights:

1.

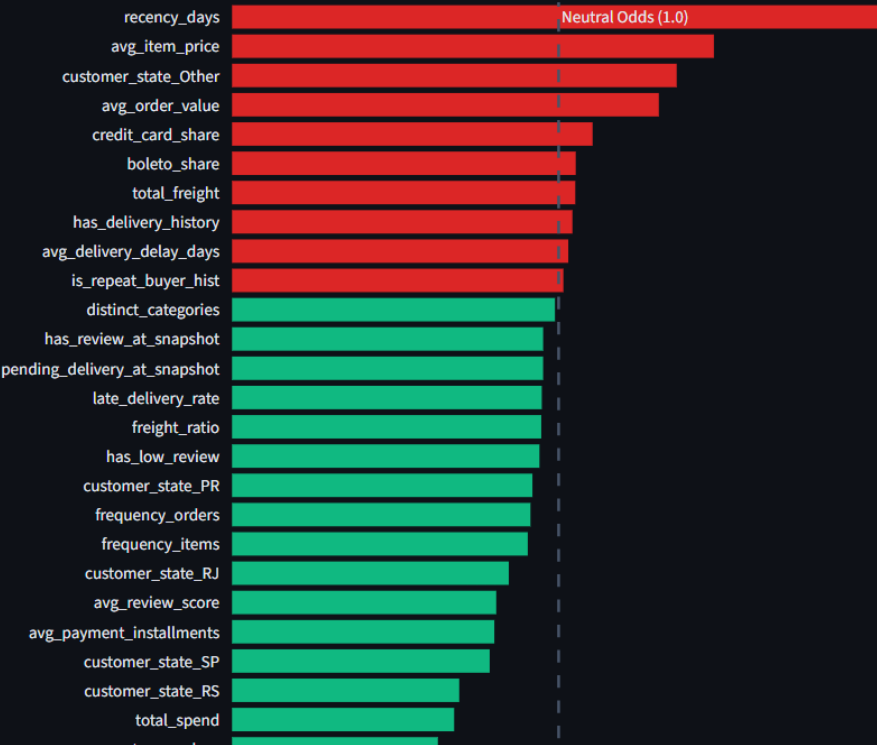
Recency is King: Every standard deviation increase in days since last purchase more than doubles churn odds ($\text{ext}\{\text{OR}\} \approx 2.08$).
2.

Tenure & Relationship: Customers with longer tenures demonstrate established loyalty ($\text{ext}\{\text{OR}\} \approx 0.58$).
3.

Review Experience: High review scores protect retention ($\text{ext}\{\text{OR}\} \approx 0.79$).
4.

Geography: Orders from São Paulo (SP) enjoy lower churn odds due to faster delivery transit

Logistic Regression Odds Ratios (e^{β})



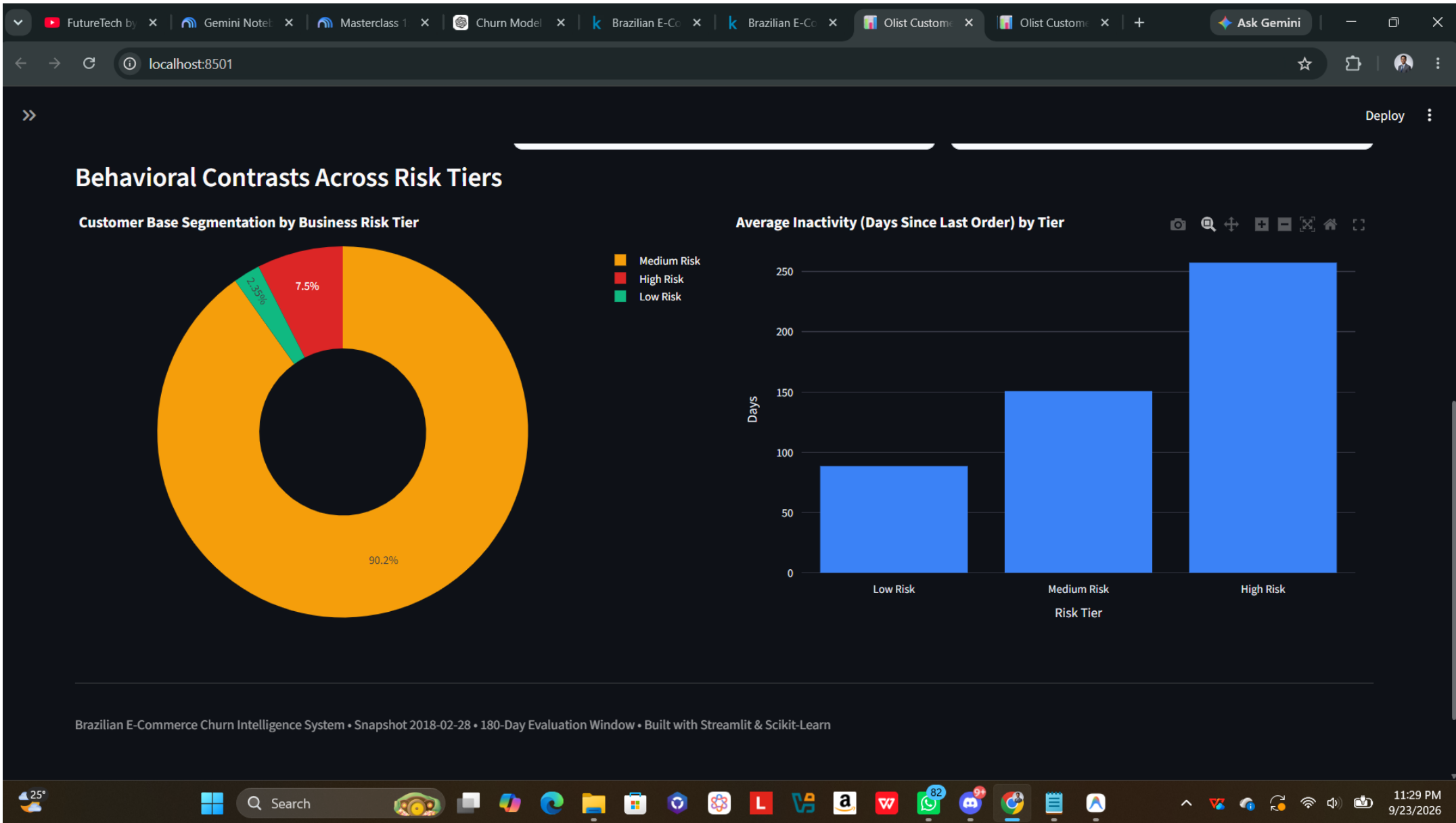
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 **Export High-Risk CSV**

Filter by Business Risk Tier:

High Risk x

Low Risk x

Filter by Historical Spend (R\$):

0.00

1000.00

	customer_unique_id	business_risk_tier	operational_risk_score	calibrated_churn_prob	risk_percentile	recency_days	tenure_days	frequenc
14	00115fc7123b5310cf6d3a3aa932699e	High Risk	76.20%	99.63%	97.4%	403	403	
15	0011805441c0d1b68b48002f1d005526	High Risk	71.02%	99.52%	93.7%	310	310	
27	002043098f10ba39a4600b6c52fbfe3c	High Risk	75.35%	99.54%	97.0%	339	339	
33	002b4cd83fabaffaa475f78ea5ef3e08	High Risk	72.65%	99.53%	95.3%	392	392	
44	0032c76b20340da25249092a268ce66c	High Risk	73.09%	99.56%	95.6%	511	511	
51	0037aa56b017c105c3a9d47af71fff98	High Risk	71.98%	99.48%	94.6%	33	33	
59	004288347e5e88a27ded2bb23747066c	Low Risk	18.03%	94.85%	0.6%	46	216	